

TechBiz Security

Cyber Security Internship Report

Week 5 – Task 15 (EH)

Brute Force & Authentication Attacks Using Hydra

Submitted By:	MANOJ
Role:	Cyber Security Intern
Internship Organisation:	TechBiz Security
Internship ID:	TBS-INT-2026-518
Submitted Date:	21 SEPT 2026
Task:	Week 5 – Task 15 (EH)
Topic:	Brute Force & Authentication Attacks Using Hydra

Table of Contents

1. Introduction
2. Objectives
3. Tools and Environment
4. Brute Force Attack Concept
5. Wordlist Verification
6. DVWA Brute Force Lab Setup
7. Burp Suite Request Interception
8. Hydra Configuration
9. Hydra Brute-Force Execution
10. Result Analysis
11. Manual Credential Verification
12. TryHackMe Hydra Room
13. TryHackMe Brute It Room
14. Defensive Measures
15. Security Impact
16. Learning Outcomes
17. Conclusion
18. Evidence / Screenshot Index

1. Introduction

This report documents the work carried out under Week 5, Task 15 (EH) of the cybersecurity internship at TechBiz Security. The task focused on understanding brute-force authentication attacks and the security risks associated with weak passwords, using Hydra as the primary testing tool.

The objective was to gain practical exposure to authentication testing methodology by performing a controlled brute-force test against the DVWA (Damn Vulnerable Web Application) Brute Force module in a local, authorized lab environment. The exercise also required completion of the TryHackMe Hydra and Brute It learning rooms to reinforce the underlying concepts.

All activities described in this report were performed strictly within a local, self-hosted lab environment set up for training purposes. No external or production systems were targeted at any point during the exercise.

2. Objectives

- Understand brute-force authentication attacks.
- Identify the risk of weak passwords.
- Locate and use a password wordlist such as rockyou.txt.
- Configure Hydra for authentication testing.
- Perform a controlled brute-force test against the DVWA Brute Force module.
- Verify the discovered credentials manually.
- Complete the required TryHackMe Hydra and Brute It learning rooms.
- Understand defensive measures against brute-force attacks.

3. Tools and Environment

The following tools and environment components were used and confirmed during the exercise:

Tool / Component	Purpose
Kali Linux	Attacker/testing operating system used for the exercise
Hydra v9.7	Network authentication brute-force tool used for the login test
DVWA (Damn Vulnerable Web Application)	Local vulnerable web application used as the authorized test target
Burp Suite Professional	Used to intercept and inspect the DVWA login request
rockyou.txt wordlist	Password wordlist supplied to Hydra for the brute-force attempt
TryHackMe Hydra room	Guided learning room on Hydra usage, completed as part of the task
TryHackMe Brute It room	Guided learning room on brute-force and password attacks, completed as part of the task
Firefox browser	Used to access DVWA locally and TryHackMe

4. Brute Force Attack Concept

A brute-force authentication attack is a method in which an attacker systematically submits a large number of username and password combinations against a login mechanism in an attempt to identify valid credentials. Rather than exploiting a software vulnerability, this technique targets weak authentication controls and predictable or commonly used passwords.

Attackers typically automate this process using tools that can send repeated login requests at speed, testing each candidate password from a prepared list against one or more usernames. Weak, default, or commonly reused passwords are particularly vulnerable because they appear early in standard password wordlists and are easily guessed within a short period.

Password wordlists, such as `rockyou.txt`, are widely used during authorized security testing because they contain large collections of passwords derived from real-world password leaks, making them effective for identifying weak credentials in a controlled assessment.

Brute-force testing of this nature must only be carried out against systems for which explicit authorization has been granted, such as a local training lab. Performing this activity against systems without authorization is illegal and unethical.

5. Wordlist Verification

The `rockyou.txt` wordlist was used as the password source for the brute-force test. The file was confirmed to be available locally at:

```
/usr/share/wordlists/rockyou.txt
```

Password wordlists such as `rockyou.txt` are useful during controlled authentication security testing because they contain a large volume of real-world, commonly used passwords. Testing an authentication mechanism against such a list helps identify accounts protected by weak or predictable passwords.

6. DVWA Brute Force Lab Setup

The practical test was carried out against the DVWA Brute Force module, hosted locally in the training environment. The vulnerability page was accessed at:

```
http://127.0.0.1:8080/DVWA/vulnerabilities/brute/
```

This confirms that the entire exercise was performed against a self-hosted, authorized local practice environment and not against any external or production system.

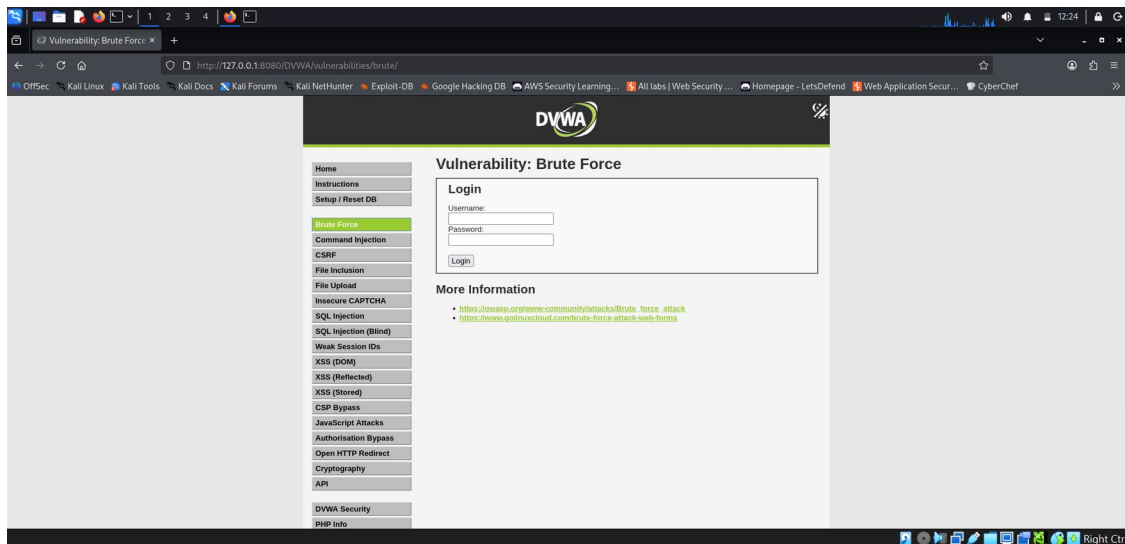


Figure 1: DVWA Brute Force vulnerability page in the local lab environment.

7. Burp Suite Request Interception

Burp Suite Professional was used to intercept and inspect the DVWA authentication request prior to configuring the Hydra HTTP form attack. The intercepted traffic confirmed the structure of the login request that Hydra would later need to replicate.

The intercepted request showed the following details:

- An HTTP GET request submitted to the DVWA Brute Force endpoint.
- Username and password parameters included in the request.
- A Login parameter confirming the form submission action.
- The target host set to the local DVWA instance (127.0.0.1).
- An authenticated/session cookie present in the request headers, required to maintain the DVWA session.

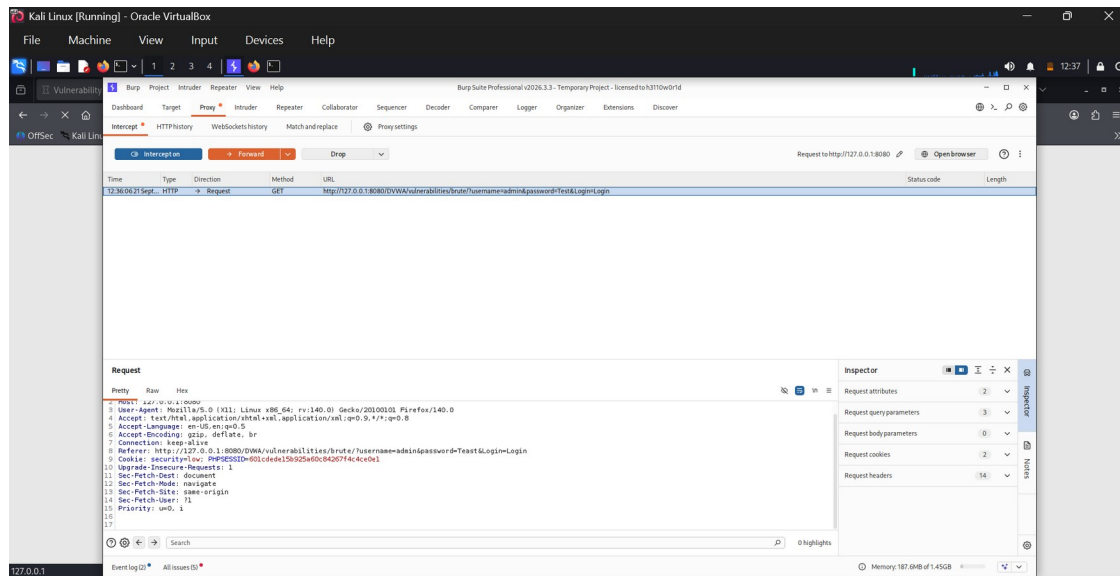


Figure 2: Burp Suite interception of the DVWA brute-force authentication request.

8. Hydra Configuration

Hydra v9.7 was used to perform the controlled brute-force test against the local DVWA application, using the request structure identified through Burp Suite. The command referenced the DVWA HTTP form login mechanism and made use of the following key components:

- Hydra: the authentication brute-force tool used to automate the login attempts.
- Username option (-l admin): specified the fixed target username, admin, based on the DVWA login form.
- Password wordlist option (-P rockyou.txt): supplied the local rockyou.txt wordlist as the source of candidate passwords.
- Local DVWA target (127.0.0.1): directed the attack at the locally hosted DVWA instance.
- Port 8080: specified the port on which the local DVWA service was running.
- HTTP GET form module (http-get-form): configured Hydra to submit login attempts using the same HTTP GET form structure observed in Burp Suite.
- DVWA authentication parameters: the username, password, and Login fields matching the DVWA Brute Force form.
- Cookie/session requirement: the request required a valid session cookie, referenced in the command as [REDACTED_SESSION_COOKIE], for the login attempt to be processed correctly by DVWA.
- Failure condition (-f): defined the text string returned by DVWA on an unsuccessful login attempt, allowing Hydra to distinguish failed attempts from a successful one.
- Controlled task count/concurrency (-t 4): limited the attack to four concurrent tasks to keep the test controlled.
- Stop-after-success behaviour: Hydra was configured to stop once a valid credential pair was identified, rather than continuing through the remainder of the wordlist.

The purpose of documenting these options is to explain how Hydra was configured to interact correctly with the DVWA authentication form, rather than to present the raw command in isolation. Any session cookie value referenced in this report has been redacted and replaced with [REDACTED_SESSION_COOKIE].

9. Hydra Brute-Force Execution

The Hydra brute-force test was executed against the local DVWA Brute Force login form. The tool output confirmed the following details of the execution:

- Tool version: Hydra v9.7.
- Target: the local DVWA service at 127.0.0.1, port 8080.
- Username tested: admin.
- Password source: the rockyou.txt wordlist.
- Concurrency: 4 tasks.
- Result: the attack completed successfully, with one valid password identified.

The valid credential pair identified by Hydra was:

Username: admin Password: password

```

manuara@kali:~$ hydra -l admin -P /usr/share/wordlists/rockyou.txt 127.0.0.1 -s 8080 http-get-form "/DVWA/vulnerabilities/brute/" "username='USER'&password='PASS'&login=Login(H:Cookie): security=low; PHPSESSID=681cdede150925a60c84287fac4ce0e1f=0s"
Hydra v9.7 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these *th* ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-09-21 12:41:26
[INFO][local] escape sequence \: detected in module option, no parameter verification is performed.
[WARNING] Hostinfofile ignored (---) from a previous session found, to prevent overwriting, ./hydra.restore
[DATA] max 4 tasks per 1 server, overall 4 tasks, 14344399 login tries (1119:14344399), ~3561800 tries per task
[DATA] attacking http-get-form://127.0.0.1:8080/DVWA/vulnerabilities/brute/"username='USER'&password='PASS'&login=Login(H:Cookie): security=low; PHPSESSID=681cdede150925a60c84287fac4ce0e1f=0s
[8000] http-get-form host: 127.0.0.1 login: admin password: password
[WARNING] attack finished for 127.0.0.1 (valid pair found)
1 of 1 target successfully completed: 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-09-21 12:41:28
manuara@kali:~$

```

Figure 3: Hydra v9.7 successfully identified a valid DVWA credential in the authorized local lab.

10. Result Analysis

The test demonstrated that the DVWA account tested accepted a weak, commonly used password, and that automated password guessing using Hydra was able to identify this credential within the controlled lab environment.

This result has the following security significance:

- Weak passwords can be discovered relatively quickly through automated password guessing.
- Automated tools such as Hydra can test large numbers of candidate passwords efficiently against a login form.
- Authentication endpoints that lack adequate protections, such as rate limiting or account lockout, can be vulnerable to repeated login attempts.
- Strong password policies and authentication controls significantly reduce the risk of successful brute-force attacks.

This result reflects the behaviour of a deliberately vulnerable training application (DVWA) in a controlled lab setting and does not represent a real-world production system compromise.

11. Manual Credential Verification

The credential pair identified by Hydra was manually tested against the DVWA Brute Force login page to confirm the result. The following credential was submitted:

Username: admin Password: password

Upon submission, the DVWA application displayed the message “Welcome to the password protected area admin”, confirming that the credential identified during the authorized lab test was accepted by the application.

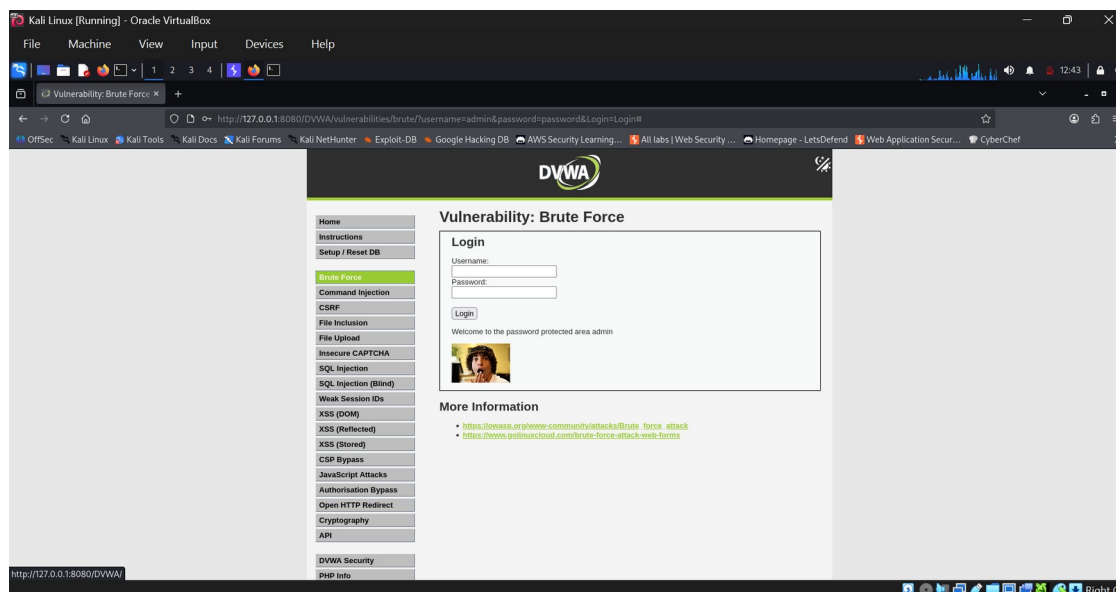


Figure 4: Manual verification of the credential identified by Hydra in DVWA.

12. TryHackMe Hydra Room

The TryHackMe Hydra room was completed as part of this task. The room provided guided, practical learning related to the Hydra tool and network login/password testing, reinforcing the concepts applied during the DVWA exercise.

The provided evidence confirms that the Hydra room was completed to 100%.

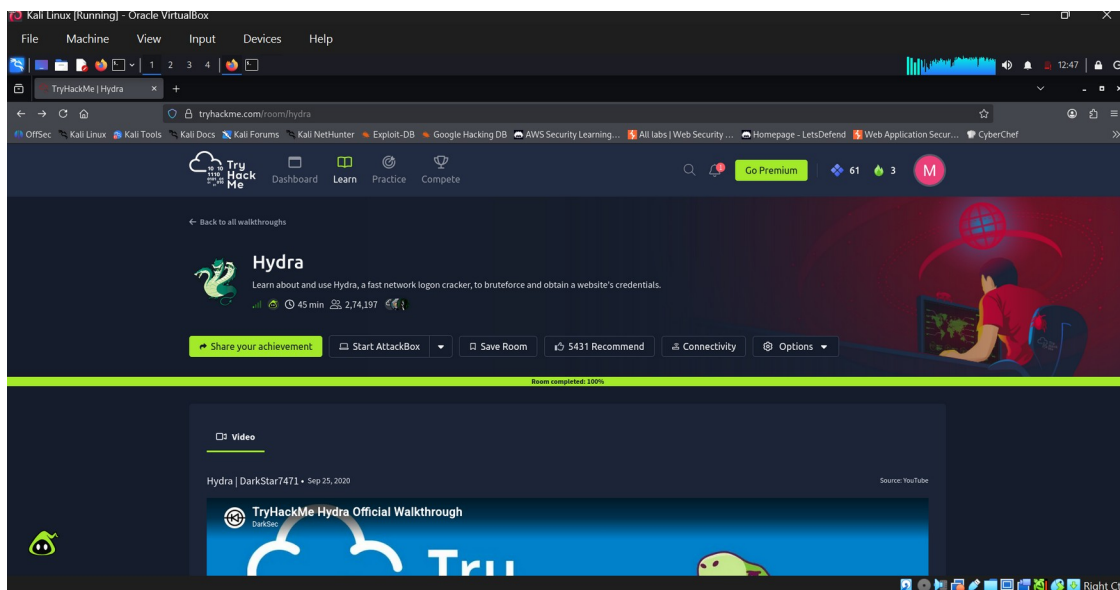


Figure 5: TryHackMe Hydra room completed at 100%.

13. TryHackMe Brute It Room

The TryHackMe Brute It room was also completed as part of this task. This room provided practical learning related to brute-force attacks, password and hash cracking, and privilege escalation concepts.

The provided evidence confirms that the Brute It room was completed to 100%.

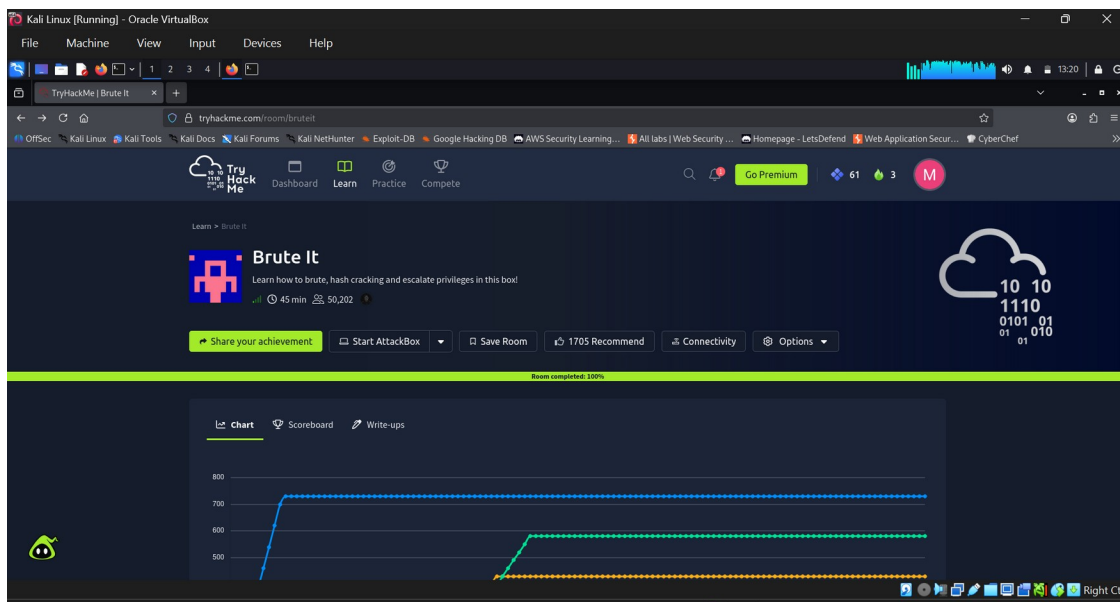


Figure 6: TryHackMe Brute It room completed at 100%.

14. Defensive Measures

Organizations can reduce the risk of brute-force authentication attacks by implementing a layered set of defensive controls rather than relying on any single measure. Recommended controls include:

- Enforcing strong password policies across all accounts.
- Requiring long and unique passwords for each user.
- Implementing multi-factor authentication (MFA).
- Applying account lockout or temporary suspension after repeated failed login attempts.
- Implementing rate limiting on authentication endpoints.
- Applying login attempt throttling to slow down repeated attempts.
- Using CAPTCHA mechanisms where appropriate.
- Monitoring authentication failures for unusual patterns.
- Deploying SIEM-based alerting for suspicious login activity.
- Detecting and flagging repeated login attempts from the same source.
- Applying IP reputation checks or blocking where appropriate.
- Storing passwords securely using strong password hashing algorithms.
- Avoiding default and commonly used passwords on all accounts.
- Conducting regular authentication security testing.

A layered defensive approach, combining several of these controls, provides significantly stronger protection against brute-force attacks than relying on a single measure.

15. Security Impact

Weak authentication controls can result in the following potential impacts:

- Unauthorized access to user accounts.
- Compromise of account credentials.
- Full account takeover by an attacker.
- Exposure of information protected behind the compromised account.
- Further compromise of connected systems if the account holds elevated privileges.

16. Learning Outcomes

Through this task, the following practical skills and concepts were developed:

- Understanding of brute-force authentication attacks and their mechanics.
- Practical use of Hydra for controlled authentication testing.
- Working with password wordlists, including rockyou.txt.
- Understanding of HTTP form authentication parameters.
- Use of Burp Suite to inspect and analyze authentication requests.
- Manual validation of security-test results.
- Understanding of the importance of strong authentication controls.
- Completion of the TryHackMe Hydra and Brute It practical learning rooms.

17. Conclusion

This task demonstrated the security risk associated with weak authentication credentials through an authorized local DVWA laboratory exercise. Using Hydra, a valid but weak credential pair was successfully identified, and the result was subsequently verified through manual login testing on the DVWA application.

In addition to the practical lab exercise, the TryHackMe Hydra and Brute It rooms were completed at 100%, providing further structured exposure to brute-force testing concepts and reinforcing the skills applied during this task.

The exercise confirms the importance of strong password policies and layered authentication defenses in protecting systems against automated password guessing attacks. All testing was conducted strictly within an authorized local training environment.

18. Evidence / Screenshot Index

No.	Screenshot	Description
1	01_dvwa_brute_force_page.png	DVWA Brute Force vulnerability page
2	02_burp_dvwa_brute_force_login_request.png	Burp Suite intercepted authentication request
3	03_hydra_bruteforce_success_admin_password.png	Hydra successful credential discovery
4	04_dvwa_brute_force_successful_login_.png	Manual DVWA credential verification
5	05_tryhackme_hydra_room_completed_100_percent.png	TryHackMe Hydra room completion
6	06_tryhackme_brute_it_room_completed_100_percent.png	TryHackMe Brute It room completion